

Body Fluids and Circulation

Topic - Full Chapter

1. Given below are two statements :

Statement-I: Systemic heart refers to left atrium and left ventricle.

Statement-II: Systemic circulation provides nutrients, O₂ and other essential substances to the tissues.

- (1) Statement I and Statement II both are correct.
- (2) Statement I is correct, but Statement II is incorrect.
- (3) Statement I is incorrect, but Statement II is correct.
- (4) Statement I and Statement II both are incorrect.

2. What is correct order of these events?

- 1. Conversion of fibrinogen of fibrin
- 2. Clot retraction and leakage of serum
- 3. Thromboplastin formation
- 4. Conversion of prothrombin to thrombin

- (1) 3,2,1,4
- (2) 3,4,1,2
- (3) 3,4,2,1
- (4) 4,1,3,2

3. Which chamber of the heart receives oxygenated blood from the lungs?

- (1) Right atrium
- (2) Left atrium
- (3) Right ventricle
- (4) Left ventricle

4. A red blood cell, entering the right side of the heart passes by or through the following structures

- 1. Atrioventricular valves
- 2. Semi-lunar valves
- 3. Right atrium
- 4. Right ventricle
- 5. SAN

- (1) 2→3→1→4→5
- (2) 3→1→5→2→4
- (3) 3→5→1→2→4
- (4) 5→3→1→4→2

5. Match List-I with List-II

List-I	List-II
I. Fibrinogen	A. Osmotic balance
II. Globulin	B. Blood clotting
III. Albumin	C. Defense mechanism
IV. Proteins	D. 6-8% of total plasma

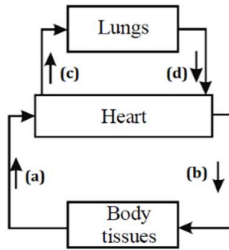
- (1) I-A, II-C, III-B, IV-D
- (2) I-A, II-B, III-C, IV-D
- (3) I-C, II-B, III-A, IV-D
- (4) I-B, II-C, III-A, IV-D

6. An open circulatory system is found in;

- (1) Reptiles
- (2) Birds
- (3) Insects
- (4) Annelids



7. The given figure represents the pathway of blood throughout the body



Identify correct of marked a , b , c and d-

- (1) a – artery
- (2) b – pulmonary vein
- (3) c – pulmonary artery
- (4) d – systemic vein

8. For safe blood transfusions :-

- (1) Donors RBC should not contain antibodies against recipient serum
- (2) Recipient serum should not contain antigens against donors antibodies
- (3) Recipient serum should not contain antibodies against donors RBCs
- (4) Recipient RBCs should not contain antibodies against donors antigens

9. Given below are two statements :

Statement-I: Arteries are thick-walled and have narrow lumen as compared to veins.

Statement-II: All veins carry deoxygenated blood except pulmonary vein.

- (1) Statement I and Statement II both are correct.
- (2) Statement I is correct, but Statement II is incorrect.
- (3) Statement is incorrect, but Statement II is correct.
- (4) Statement I and Statement II both are incorrect.

10. Given below are two statements :

Assertion (A) : The end of T-wave marks the end of ventricular systole.

Reason (R) : T-wave represents the return of ventricles from excited to normal state which is repolarization.

- (1) Both Assertion (A) and Reason (R) are true, and Reason (R) is a correct explanation of Assertion (A).
- (2) Both Assertion (A) and Reason (R) are true, but Reason (R) is not a correct explanation of Assertion (A).
- (3) Assertion (A) is true, and Reason (R) is false.
- (4) Assertion (A) is false, and Reason (R) is true.

11. Which of the following statements are incorrect?

- (1) among all the four chambers of heart left ventricle has the thickest walls
- (2) contraction of the ventricle in the heart begins by the command from SA node
- (3) QRS complex in ECG is related with ventricular contraction and atrial relaxation
- (4) human heart is myogenic

12. If due to some injury the chordae tendineae of the tricuspid valve of the human heart is partially nonfunctional, what will be the immediate effect?

- (1) The flow of blood into the aorta will be slowed down.
- (2) The pacemaker will stop working.
- (3) The blood will tend to flow back into the left atrium.
- (4) The flow of blood into the pulmonary artery will be reduced.

13. Given below is ECG of a normal human. Which one of the following component is correctly interpreted below?



- (1) Peak P and peak R represent systolic and diastolic blood pressures respectively.
- (2) Peak P - Initiation of left atrial contraction only.
- (3) Complex QRS - One complete pulse.
- (4) Peak T - Initiation of total cardiac contraction.

14. Consider the following statements

Statement-I: When a person is performing normal work. On an average, his heartbeat is 72-75 heart beats per minute.

Statement-II: One cardiac cycle is completed in 0.8 second.

- (1) Statement I and Statement II both are correct.
- (2) Statement I is correct, but Statement II is incorrect.
- (3) Statement I is incorrect, but Statement II is correct.
- (4) Statement I and Statement II both are incorrect.

15. Which hormone decreases heart beat rate thereby decreases the cardiac output

- (1) Thyroxin
- (2) Sympathin
- (3) Acetylcholine
- (4) Adrenaline

16. Which of the following events do not occur during joint diastole?

- I. All 4 chambers of heart are in relaxed state**
- II. Tricuspid and bicuspid valves open**
- III. Action potential is conducted from SAN to AVN**
- IV. Blood from the pulmonary veins and vena cava flows into the left and right ventricles respectively through the left and right atria.**
- V. The semilunar valves are closed.**

- (1) Only V
- (2) Only III
- (3) Only IV
- (4) Only I and II

17. Match List-I with List-II

List-I	List-II
I. Ventricular systole	A. 0.4 second
II. Ventricular diastole	B. 0.1 second
III. Atrial systole	C. 0.3 second
IV. Joint diastole	D. 0.5 second

- (1) I-A, II-B, III-D, IV-C
- (2) I-C, II-D, III-A, IV-B
- (3) I-C, II-D, III-B, IV-A
- (4) I-D, II-C, III-A, IV-B

18. In a normal man blood pressure is;

- (1) 120/80 mm of Hg
- (2) 80/100 mm of Hg
- (3) 80/120 mm of Hg
- (4) 100/80 mm of Hg

19. Given below are two statements :

Assertion (A) : SA node malfunction leads to disturbance of heart rate.

Reason (R) : SA node is the pacemaker of heart producing electric impulse for heart contraction.

(1) Both Assertion (A) and Reason (R) are true, and Reason (R) is a correct explanation of Assertion (A).

(2) Both Assertion (A) and Reason (R) are true, but Reason (R) is not a correct explanation of Assertion (A).

(3) Assertion (A) is true, and Reason (R) is false.

(4) Assertion (A) is false, and Reason (R) is true.

20. Match List-I with List-II to find out the correct option.

List-I

List-II

I. Fishes

A. Single circulation

II. Amphibians

B. Incomplete double circulation

III. Reptiles

C. Double circulation

IV. Birds

V. Mammals

(1) I-A, II-A, III-B, IV-B, V-C

(2) I-A, II-A, III-B, IV-C, V-C

(3) I-A, II-B, III-B, IV-C, V-C

(4) I-A, II-B, III-C, IV-B, V-B

ANSWER KEY

1. (3) 2. (2) 3. (2) 4. (4) 5. (4) 6. (3) 7. (3) 8. (3) 9. (1) 10. (1)
11. (3) 12. (4) 13. (3) 14. (1) 15. (3) 16. (2) 17. (3) 18. (1) 19. (1) 20. (3)

